Longsnout seahorse

A tail that holds on: the biomechanics behind the Planted Tank's seahorses

Hippocampus reidi · World Foundry aquarium series

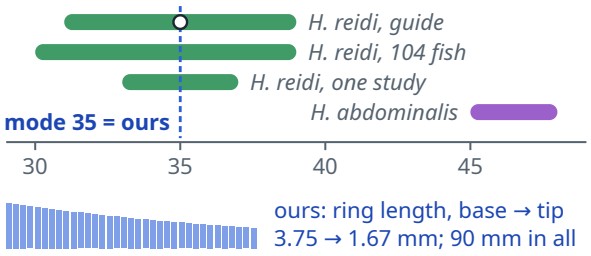
verified Captive breeding pairs average 15.6–16.6 cm in height; the largest adult on record is 17.5 cm. [1, 2]

verified Colour varies: brown spots with tiny white dots, striped rings, or one even colour. [2, 3]

verified Western Atlantic; on seagrass, mangrove roots, macroalgae and gorgonians. [2]

01 · Thirty-five rings





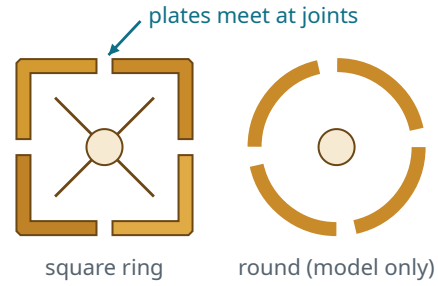
verified 11 trunk rings, and most often 35 tail rings (range 31–39). [2]

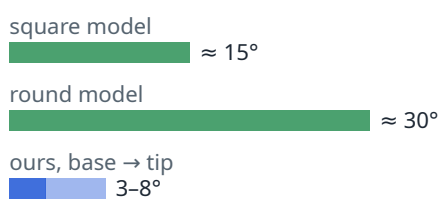
verified Other counts: 30–39 tail rings in 104 Brazilian specimens; 33–37 in another study. [3, 4]

verified Each tail ring is four L-shaped bony plates around one vertebra. [5]

ours Our tail: 35 rings, 90 mm. Rings shrink from 3.75 to 1.67 mm long and 9.0 to 1.4 mm wide.

02 · Square bends without crushing



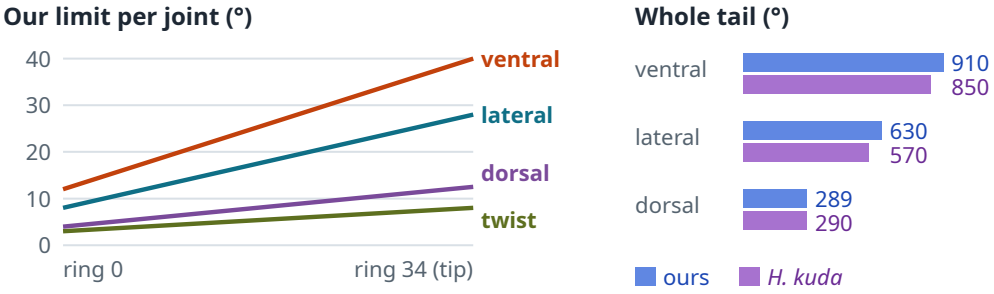


verified Joints: ball-and-socket between vertebrae, peg-and-socket between plates, gliding where plates overlap. [5]

verified In 3D-printed models, the square tail resists crushing better than a round one, and it twists about 15° per segment (round: about 30°). [5]

ours Our twist limit rises from 3° at the base to 8° at the tip.

03 · More motion toward the tip



verified *H. reidi* grips a 1 cm perch with the whole tail, with more range of motion toward the tip. The tail also bends sideways as it curls. [4]

other species *H. kuda*: the whole tail bends about 850° ventrally, 290° dorsally and 570° sideways. [5]

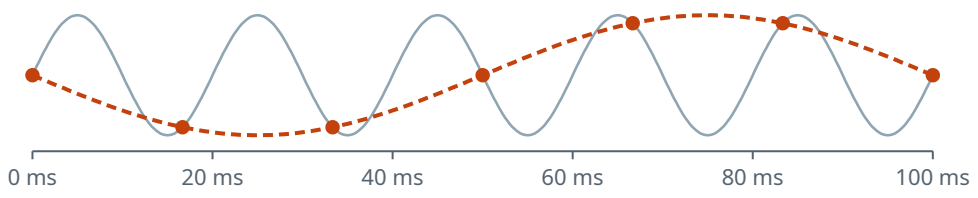
other species *H. abdominalis* chose vertical, round, smooth holdfasts. [4]

ours Our limits rise linearly toward the tip; our wrap closes from the tip.

04 · A fin too fast to draw

A 50 Hz fin beat, seen by a 60 Hz screen

Red dots: one per 16.7 ms frame. Joined up, they look like a 10 Hz wave.



At 13.5 FPS, Chromecast 1 draws a frame only every 74 ms.

verified *H. reidi* fins: 20–60 Hz, slowest when hovering; most often about 50 Hz. [6, 7]

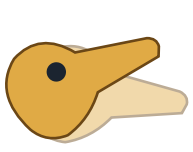
other species *H. erectus* (as *H. hudsonius*) dorsal fin: 30–42 Hz; dwarf seahorse: 42–54 Hz. [8]

other species *H. kuda* holds at most 3–4 cm/s against a current. [9]

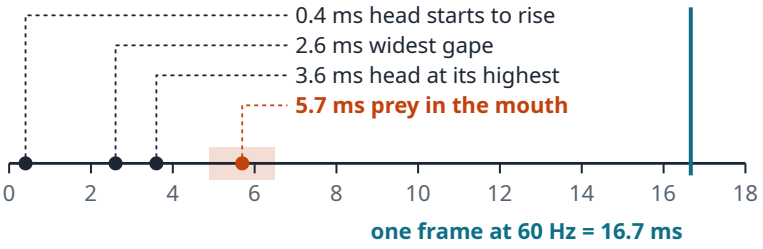
ours A 60 Hz screen shows at most 30 Hz, so we draw a slowed, blurred wave. Swim speed is ours.

05 · A six-millisecond strike

Adult strike, timed from the first hyoid movement (ms)



peak head rotation
26.7 ± 8° [10]



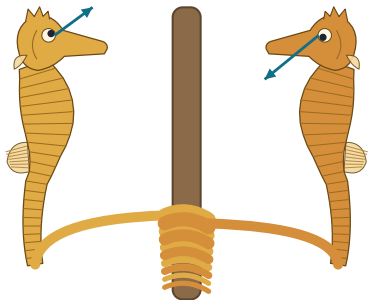
verified Adult *H. reidi* pivot feeding: the head starts to rise 0.4 ms after the hyoid moves; the prey crosses the mouth at 5.7 ± 0.8 ms. [10]

verified Newborns turn the head more than 40° in under 2.5 ms. [11]

ours The strike ends inside one 16.7 ms frame at 60 Hz, so the game cuts from aim to strike.

06 · Pairs, and eyes that work alone

Each eye can track its own target



one shared post

verified *H. reidi* is found in pairs in the wild. [2]

other species Seahorses and pipefishes move each eye on its own to find, track and strike single prey. [12]

ours Our morning greeting, two tails on one post, and a separate target for each eye are game choreography.

07 · From rings to joints

35 rings, one joint each; the large dots are the 7 knots



knots **7 × 3 = 21** + wrap **4 values** → from the script **25 values** → engine output **35 × 3 = 105 angles**

mailbox block: 40 cells

ours One joint per ring, each with flex, sideways bend and twist: 35 × 3 = 105 angles.

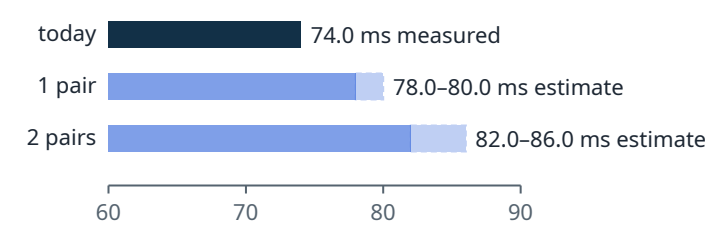
ours The script writes 25 values: 7 knots × 3 angles, plus a 4-value wrap. The engine fills in every joint and clamps each to its limit.

ours Clip, head, eye and output cells bring the mailbox block to 40 cells.

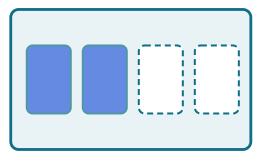
08 · What a pair costs

Chromecast 1 frame time (ms)

The axis starts at 60 ms.



Our 40 gal tank



guideline allows 4 pairs; **we stop at 2**

ours Planted tank today: 74.0 ms per frame (13.5 FPS) on Chromecast 1, measured 2026-10-05; stale, so Phase 4 measures it again.

estimate (ours) 2–3 ms per seahorse, scaled from the urchin: one pair 78.0–80.0 ms, two pairs 82.0–86.0 ms.

other species Breeder guidance for *H. erectus*: one pair of adults per 10 gal. [13]

ours Our 40 gal tank: None, 1 pair or 2 pairs.

verified source read; it says this of *H. reidi* **other species** source read; it is about the species named **ours** a game choice; unmeasured costs say **estimate (ours)**

[1] Hettiarachchi & Edirisinghe 2017. Trop. Agric. Res. 29: 77–89 · abstract · *H. reidi*

[2] Lourie, Foster, Cooper & Vincent 2004. A Guide to the Identification of Seahorses · pp. 2, 72–73 · *H. reidi*

[3] Silveira et al. 2014. Zootaxa 3861: 317–332 · pp. 322–323 · *H. reidi*

[4] Maia, Couto & Adriaens 2013. How seahorses hang on to their life. SICB abstract · abstract · *H. reidi, H. abdominalis*

[5] Porter et al. 2015. Why the seahorse tail is square. Science 349: aaa6683 · Figs. 1, 2, 5 · *H. reidi, H. kuda; printed models*

[6] Zima-Kulisiewicz et al. 2009. Undulatory fin motion of seahorse. GALA proceedings 56 · pp. 56-4–56-5 · *H. reidi*

[7] Botello-Payro et al. 2010. Flow induced by seahorse fins. GALA proceedings 28 · results · *H. reidi*

[8] Ashley-Ross 2002. J. Exp. Zool. 293: 561–577 · p. 566 · *H. hudsonius (= H. erectus), H. zosterae*

[9] Li & Xu 2016. Swimming ability of the yellow seahorse. Marine Sciences 40(5): 76–81 · abstract · *H. kuda*

[10] Roos et al. 2009. Suction feeding in H. reidi. J. Exp. Biol. 212: 3490–3498 · p. 3493, Table 1 · *H. reidi*

[11] Van Wassenbergh et al. 2009. Suction is kid's play. Biol. Lett. 5: 200–203 · p. 201, Fig. 2 · *H. reidi newborns*

[12] Mosk et al. 2007. Visual Neuroscience 24: 345–354 · abstract · *Syngnathidae*

[13] Ocean Rider (breeder). Stocking Density forum thread, 2006 · reply #2823 · *H. erectus*

Sources checked 2026-10-16. Locators, DOIs and access limits: docs/reference/seahorse-research.md. The drawings are original and use the game's own ring table. Claims the research could not confirm stay in the research document. A3 portrait, 297 × 420 mm; print on matte or uncoated stock.